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The Paradox of Social Distancing Loneliness

For over a year, humans have maintained distancing measures to prevent the spread of Covid 19. While lockdowns and distancing prevented a more considerable spread, it also prevented behaviors and interactions necessary to maintain social networks and social bonds. Many people (63% of adults, according to one national survey) experienced some degree of loneliness during that time, suggesting that with an easement of distancing would be an increase in social activity. However, another survey reported that half of Americans were feeling somewhat uneasy about resuming social activities and in-person social interactions, not necessarily due to reasons related to physical health. While it sounds like losing the ability to maintain the social bonds necessary for survival is a somewhat dire scenario, the same plasticity that allows isolation to affect the brain so intensely is the same reason the negative effects of isolation can be reversed, despite initial paradoxical feelings of hesitance and anxiety.

Since our earliest social structures, humans have relied upon social networks for purposes ranging from mating to collective protection and hunting. This phenomenon is not solely at play in human social networks – the concept of social homeostasis has been studied across species, all leading to similar conclusions. Smaller social networks provide less social benefits, lead to higher levels of the stress hormone cortisol, and an increase in anxiety-adjacent behaviors. However, big social networks lead to increased competition. Some degree of brain function has to maintain a degree of social equilibrium to regulate the balance of socialization. In humans, the mesocorticolimbic circuit and hippocampus both play roles in gauging and responding to social interactions.

The mesocorticolimbic circuit is associated with our reward system, inducing cravings and fulfillment-seeking behaviors. Its role in this experience is notable; a reduction in socialization shows an increase in social cravings and patterns of brain activity comparable to food cravings. This held true in studies using other species as model specimens. There may be an evolutionary reason for those cravings as well: in isolation, an individual experiences heightened vigilance toward threats. In social isolation, an individual experiences that same hypervigilance toward perceived social threats. No wonder then that after a period of isolation, an individual might be hesitant to get back out there.

Another region of the brain notable to this discussion is the hippocampus, where learning and memory take place. Much of the success of our social circles relies on learned and remembered behaviors. However, our brain must constantly store and integrate new and relevant information, so after a period of isolation (i.e. not using those learned and remembered behaviors), social and working memory can be impaired. A study on the hippocampi of isolated Antarctic expeditioners provided evidence supporting this; after 14 months of isolation, the hippocampi had shrunk.

Regardless of the perceived social threats, social homeostasis is critical for our survival. Thankfully, the neuroplasticity of the brain that led to those circumstances can also reverse them. Studies have suggested that resocialization, despite the initial surge of anxiety, can repair the effects of anxiety and stress associated with isolation. Our social memory and cognitive function also show a degree of adaptability upon resocialization. Studies on marmots and rodents provided evidence supporting the positive effects of resocialization both on the effects of isolation and quick regaining of memory after resocialization. Studies in regards to this specific experience are somewhat limited; in one human study – some people experienced cognitive decline during lockdown but recovered after easing of restrictions.

Unfortunately, experiments on animal in extreme laboratory settings show only one facet of the possibilities. We, as humans, were still linked socially via technology and did not experience the trauma of complete and total isolation. Studies are still few and far between, so it remains to be seen whether the long-term data holds true – in any case, diving headfirst into the initial surge of post-isolation anxiety and resuming normal socialization should reset social homeostasis and reverse some of the stressful effects of pandemic lockdown.

Clark, Kareem. “The Neuroscience Behind Why Your Brain May Need Time to Adjust to Un-Social Distancing”. *The Conversation*, 28 June 2021. <https://theconversation.com/the-neuroscience-behind-why-your-brain-may-need-time-to-adjust-to-un-social-distancing-162075>. Accessed 21 March 2022